WORKED SOLUTION

STEP III 2010, Question 5

Pure • Coordinate Geometry Construction

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s3-q5

The key idea

Coordinates do all the work here: find R and S, write down the lines PQ and RS in intercept form, and look at where each of them crosses the line through A perpendicular to AC. They cross it at the *same* point, and that point must be T. The construction then rests on one observation: the diagonal of the given square is exactly the side we need, and the first part gives us a straight-edge-only way to draw a right angle to that diagonal.

Solution

Showing that $TA \perp AC$

Lines through $C(a,a)$ are easy to handle. The line CP joins (a,a) to $(an,0)$, so its gradient is $\dfrac{a}{a-an} = \dfrac{1}{1-n}$ (defined, since $n<1$) and its equation is

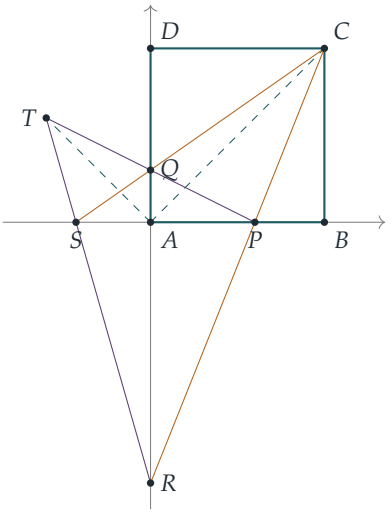
$$y = \frac{x-an}{1-n}.$$

The line DA is $x=0$, so

$$R = \left(0,\ -\frac{an}{1-n}\right).$$

Similarly CQ joins (a,a) to $(0,am)$, has gradient $1-m \neq 0$, and is $y = am + (1-m)x$. Setting $y=0$ (the line BA),

$$S = \left(-\frac{am}{1-m},\ 0\right).$$



Now use intercept form. The line PQ has intercepts an and am; the line RS has intercepts $-\frac{am}{1-m}$ and $-\frac{an}{1-n}$. So

$$\begin{aligned} PQ: &\quad \frac{x}{n}+\frac{y}{m}=a, \\ RS: &\quad -\frac{(1-m)x}{m}-\frac{(1-n)y}{n}=a. \end{aligned}$$

The line through A perpendicular to AC is $y=-x$, since AC has gradient 1. Put $y=-x$ into each equation:

$$\begin{aligned} PQ: &\quad x\Big(\frac{1}{n}-\frac{1}{m}\Big)=a, \\ RS: &\quad x\Big(-\frac{1}{m}+1+\frac{1}{n}-1\Big)=x\Big(\frac{1}{n}-\frac{1}{m}\Big)=a. \end{aligned}$$

These are the same equation. Because $m \neq n$, the bracket is non-zero, and both lines meet $y=-x$ at the single point

$$\Big(\frac{amn}{m-n},\ -\frac{amn}{m-n}\Big).$$

Ask yourself We've found a point that lies on both PQ and RS. Why must it be T?

Two distinct lines meet in at most one point. The lines PQ and RS are distinct (PQ passes through P with $x>0$ on the x-axis, while RS meets the x-axis at S with $x<0$). So any point common to both is their intersection T. Hence

$$T=\Big(\frac{amn}{m-n},\ -\frac{amn}{m-n}\Big),$$

which lies on $y=-x$. The vector $\overrightarrow{AT}$ is a multiple of $(1,-1)$ and $\overrightarrow{AC}=(a,a)$, and $(1,-1)\cdot(a,a)=0$. So TA is perpendicular to AC, as required.

Notice that the working never used $m < n$, only $m \neq n$. It also never used the actual positions of P and Q, so *any* two points P on side AB and Q on side AD, provided that $AP \neq AQ$, give a T on the perpendicular to AC at A.

The construction

A square of area $2a^2$ has side $\sqrt{2}\,a$, which is the length of the diagonal AC. So we build a square on AC as one side. In coordinates, this is the square with vertices

$$A(0,0), \quad E(a,-a), \quad F(2a,0), \quad C(a,a).$$

Its sides $\overrightarrow{AE} = (a,-a)$, $\overrightarrow{EF} = (a,a)$, $\overrightarrow{FC} = (-a,a)$ and $\overrightarrow{CA} = (-a,-a)$ all have length $\sqrt{2}\,a$, and consecutive sides are perpendicular, so its area is $2a^2$.

Ask yourself With an unmarked ruler we can only join points and extend lines. How can we possibly get a right angle?

The first part is exactly such a recipe, since every step in it joins two known points or intersects two lines. With the straight-edge:

1. Draw the diagonal AC. Mark any point P on AB and any point Q on AD, choosing them visibly unequal distances from A (no measurement is needed; we only need $AP \neq AQ$).
2. Extend CP to meet line DA at R, and extend CQ to meet line BA at S. Extend PQ and RS to meet at T.
3. Draw line TA. By the result above it is perpendicular to AC. Extend CB beyond B to meet line TA at E. (Line CB is $x = a$ and line TA is $y = -x$, so $E = (a,-a)$.)
4. Repeat steps 1–3 at the opposite corner C: mark points on CD and CB at visibly unequal distances from C and run the same construction. The half-turn about the centre of the square swaps A and C and maps the square to itself, so this gives the line through C perpendicular to AC, namely $x + y = 2a$.
5. Extend AB beyond B to meet this line at $F = (2a,0)$.

Then $AEFC$ is a square of side $AC = \sqrt{2}\,a$, so its area is $2a^2$. (Its diagonals AF and CE lie along lines AB and CB and cross at B, the centre of the new square.)

Checks

- Take $a = 1$, $m = 0.3$, $n = 0.6$. Then $R = (0,-1.5)$, $S = (-\frac{3}{7},0)$ and $T = (-0.6,0.6)$. The point T lies on the line through R and S: $-\frac{0.7}{0.3}(-0.6) - \frac{0.4}{0.6}(0.6) = 1.4 - 0.4 = 1$. It also lies on PQ: $-1 + 2 = 1$. And T is on $y = -x$.
- By computer we computed R, S and T exactly, using fractions, for 2000 random choices of a, m and n, in both orders $m < n$ and $m > n$. In every case $\overrightarrow{AT} \cdot \overrightarrow{AC} = 0$,

and the constructed $AEFC$ had four equal sides of squared length $2a^2$ meeting at right angles.

Common slips

- Solving PQ and RS simultaneously by brute force. It works, but it's messy. Testing a candidate line ($y = -x$) against both equations is far quicker.
- Forgetting to explain why the common point really is T. You need the lines to be distinct.
- In the construction, "drawing a perpendicular" or "marking off a length" without saying how. Every step must be joining points or intersecting lines.

Beyond STEP

Constructions using only a straight-edge belong to *projective geometry*, where joining and intersecting are the only operations, and ratios like the ones hidden in R, S and T are studied through cross-ratios and harmonic ranges. A classical theorem of Poncelet and Steiner says that a straight-edge alone can do everything a ruler and compass can, once a single circle and its centre are drawn; here the square plays a similar role. Coxeter's *Projective Geometry* is a beautiful introduction to the subject.

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